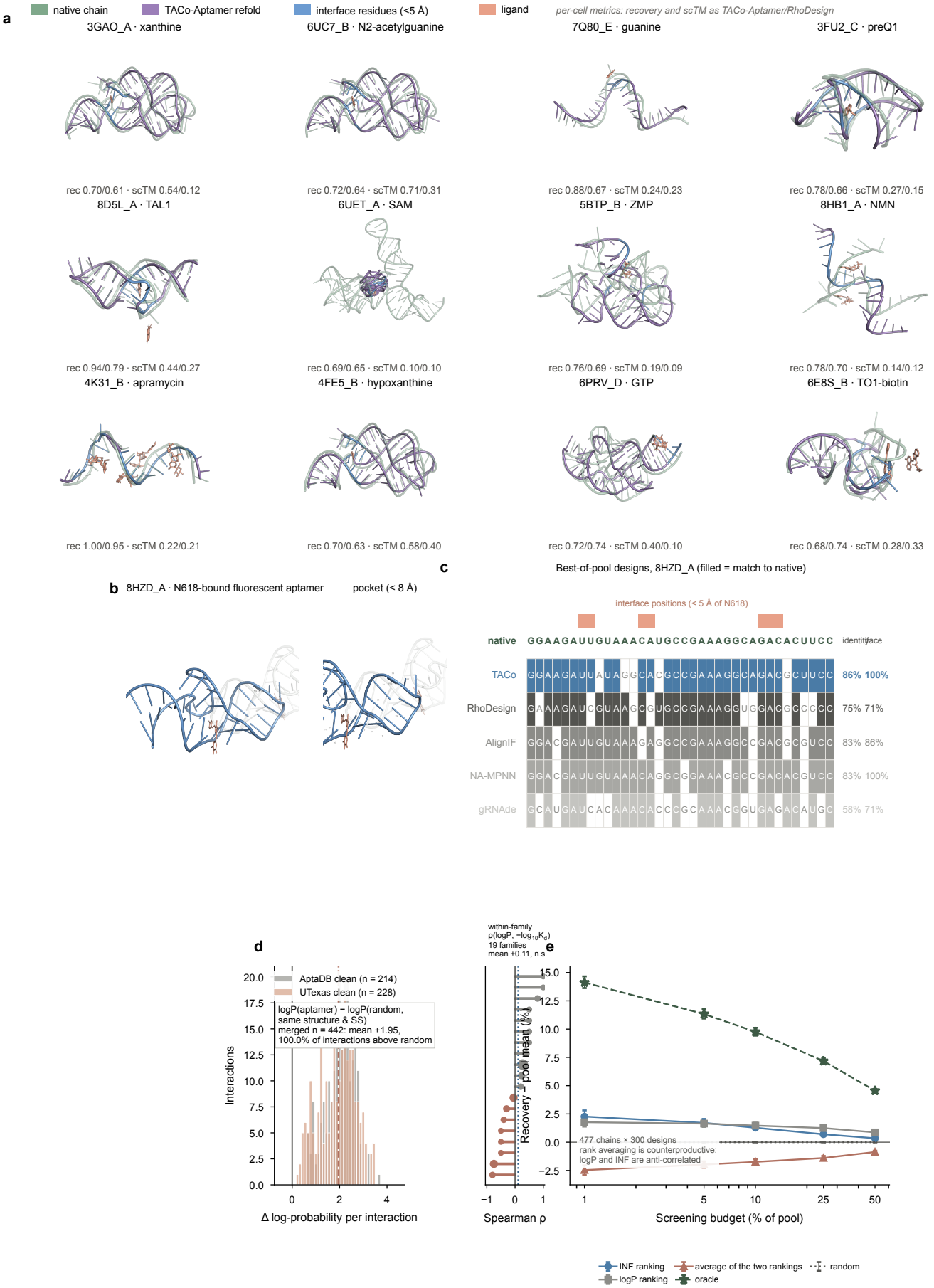

a
native chain
TACo-Aptamer refold
interface residues (<5 Å)
ligand
per-cell metrics: recovery and scTM as TACo-Aptamer/RhoDesign
3GAO_A · xanthine
rec 0.70/0.61 · scTM 0.54/0.12
6UC7_B · N2-acetylguanine
rec 0.72/0.64 · scTM 0.71/0.31
7Q80_E · guanine
rec 0.88/0.67 · scTM 0.24/0.23
3FU2_C · preQ1
rec 0.78/0.66 · scTM 0.27/0.15
8D5L_A · TAL1
rec 0.94/0.79 · scTM 0.44/0.27
6UET_A · SAM
rec 0.69/0.65 · scTM 0.10/0.10
5BTP_B · ZMP
rec 0.76/0.69 · scTM 0.19/0.09
8HB1_A · NMN
rec 0.78/0.70 · scTM 0.14/0.12
4K31_B · apramycin
rec 1.00/0.95 · scTM 0.22/0.21
4FE5_B · hypoxanthine
rec 0.70/0.63 · scTM 0.58/0.40
6PRV_D · GTP
rec 0.72/0.74 · scTM 0.40/0.10
6E8S_B · TO1-biotin
rec 0.68/0.74 · scTM 0.28/0.33
b
8HZD_A · N618-bound fluorescent aptamer
pocket (< 8 Å)
c
Best-of-pool designs, 8HZD_A (filled = match to native)
interface positions (< 5 Å of N618)
native GGAAGAUUGUAAACAUGCCGAAAGGCAGACACUUCC
identity
face
TACo GGAAGAUUAUAGGCACGCCGAAAGGCAGACGCUUCC 86% 100%
RhoDesign GAAAGAUCGUAAGCGUGCCGAAAGGUGGACGCCCCC 75% 71%
AlignIF GGACGAUUGUAAAGAGGCCGAAAGGCCGACGCGUCC 83% 86%
NA-MPNN GGACGAUUGUAAACAGGCGGAAACGCCGACACGUCC 83% 100%
gRNAde GCAUGAUCACAAACACCCGCAAACCGUGAGACAUGC 58% 71%
d
AptaDB clean (n = 214)
UTexas clean (n = 228)
logP(aptamer) − logP(random, same structure & SS)
merged n = 442: mean +1.95,
100.0% of interactions above random
Interactions
Δ log-probability per interaction
e
within-family
ρ(logP, −log10Kd)
19 families
mean +0.11, n.s.
Recovery of pool mean (%)
477 chains × 300 designs
rank averaging is counterproductive:
logP and INF are anti-correlated
Spearman ρ
Screening budget (% of pool)
INF ranking
logP ranking
average of the two rankings
oracle
random